# LTA analysis - Descriptive Stats

Annie Green Howard

### Reading in dataset

Reading in the combined years datasets

```
combine_yrs <- read.table("Data/lca_combined_years.dat",
                          header = TRUE, sep = ",", skip = 2)

#Changing all values of -9999 to missing
combine_yrs[combine_yrs==-9999] <- NA
```

### Adding in variable names to file

There appear to be a total of 16,947 observations with 78 variables.

It's not completely clear in the published paper there were 8,702 individuals in 2009 and 7,929 in 2019 - for a total sample size of 16,631 - but this doesn't seem to allow me to link the same individuals over time.

### Recreating descriptive tables

We could potentially assume group is year given group = 1 for 7,929 individuals, which matches the sample size in published paper for 2019. However, there are a total of 9,018 with a value of group = 0 - which is larger than the 8,702 individuals in the descriptive table for 2009.

```
table(combine_yrs$group)
```

```
   0    1
9018 7929
```

### Looking at the 2009 dataset and getting it to be similar to 2019 dataset

The 2009 dataset does include closer to the true number, including 8,699 individuals. Same sample size for both 1.24.23 version and the v2 version. These datasets both include indexno variable. We cannot seem to get to exactly 8,702 individuals, in Table 1 in published paper, was there exclusion criteria we care about there where we might also want to include them in the descriptive table for LTA work?

```
#Reading in the version 2 datasets for 2009
##but this seems to be an older version, dated 11/22/2023
lca_pcp_2009v2      <- read.table("Data/lca_pcp_2009v2.dat",
                                  header = TRUE, sep = ",", skip = 2)
#Changing all values of -9999 to missing
lca_pcp_2009v2[lca_pcp_2009v2==-9999] <- NA

#Reading in the datasets for 2009 and 2019 as of 1.24.23
lca_pcp_2009_12423  <- read.table("Data/lca_pcp_2009_1.24.23.dat",
                                  header = TRUE, sep = ",", skip = 2)
#Changing all values of -9999 to missing
lca_pcp_2009_12423[lca_pcp_2009_12423==-9999] <- NA
```

Now we need to format the variables to be categorical variables

```
lca_pcp_2009_12423 <- lca_pcp_2009_12423 %>%
  mutate(specialty = case_when(
    group_fam   == 1 ~ "Family Medicine",
    group_int   == 1 ~ "General Internal Medicine",
    group_obgyn == 1 ~ "General Obstetrics and Gynecology",
    group_ped   == 1 ~ "General Pediatrics",
    group_ger   == 1 ~ "Geriatrics",
    TRUE             ~ "Missing"
  ),
  employment_setting = case_when(
    setting_amb    == 1 ~ "Ambulatory Care",
    setting_hosp   == 1 ~ "Hospital",
    setting_medsch == 1 ~ "Medical School/Parent University",
    setting_nurs   == 1 ~ "Nursing Home/Extended Care",
    setting_other == 1 ~ "Other",
    setting_solo == 1 ~ "Solo practioner",
    setting_missing == 1 ~ "Missing",
    TRUE             ~ "Missing Value"
  ),
```

```r
  urban_rural = case_when(
    ruca_urban  == 1 ~ "Urban",
    ruca_lrural == 1 ~ "Large Rural",
    ruca_srural == 1 ~ "Small Rural",
    TRUE             ~ "Missing"
  ),
  race_ethnicity = case_when(
    white == 1           ~ "Non-Hispanic White",
    black == 1           ~ "Non-Hispanic Black",
    american_native == 1 ~ "American Indian/Alaskan Native",
    asian_or_pi == 1     ~ "Asian/Pacific Islander",
    hispanicrace == 1    ~ "Hispanic",
    other_race == 1      ~ "Other",
    race_unknown == 1    ~ "Unknown",
    TRUE                 ~ "Missing"
  ),
  generational = case_when(
    gen_x          == 1 ~ "Gen X (1965-1980)",
    gen_y          == 1 ~ "Gen Y (1981-2000)",
    gen_boom_lead  == 1 ~ "Baby Boomer, Leading Edge (1946-1954)",
    gen_boom_trail == 1 ~ "Baby Boomer, Trailing Edge (1955-1964)",
    gen_ww2        == 1 ~ "WWII (1928-1945)",
    TRUE                ~ "Missing"
  ))
```

### Specialty

We do note that 3 individuals, out of the 8,699 are missing specialty. Although none are mentioned in Table 1 as missing specialty.

```r
table(lca_pcp_2009_12423$specialty)
```

```
                  Family Medicine         General Internal Medicine
                             2826                              3071
General Obstetrics and Gynecology               General Pediatrics
                             1127                              1630
                       Geriatrics                           Missing
                               42                                 3
```

### Employment Setting

We do have 550 individuals missing employment setting which aligns with Table 1, where we see some degree of missingness for employment setting. Do we drop these individuals from the analysis? We could leave them in but there are two different approaches - it's not super clear what was used before but we could essentially use maximum likelihood methods - leave them as missing - or create missingness categories. I think it depends on why think people are missing.

```
table(lca_pcp_2009_12423$employment_setting)
```

```
               Ambulatory Care                       Hospital
                          4604                           1750
Medical School/Parent University                      Missing
                           648                            550
      Nursing Home/Extended Care                        Other
                            71                            113
               Solo practioner
                           963
```

### Practice Characteristics

While other variables are mututally exclusive providing routine obstetric care and providing routine prenatal care appear to be not mutually exclusive.

```
table(lca_pcp_2009_12423$prenatal,lca_pcp_2009_12423$obstetrics)
```

```
       0    1
  0 7060   20
  1  262  848
```

### Average patient-care hours per week

Check on number of missing values for average patient-care hours per week - we have a total of 626 individuals missing.

```
sum(is.na(lca_pcp_2009_12423$patient_care_hours))
```

```
[1] 626
```

Looking at descriptive statistics by patient care

```r
#Mean for patient care hours
mean(lca_pcp_2009_12423$patient_care_hours, na.rm=TRUE)
```

```
[1] 40.71436
```

```r
#Mean for Standardized values of Y2009_patient_care_hours
mean(lca_pcp_2009_12423$zhours, na.rm=TRUE)
```

```
[1] 0.0004237322
```

**Urban/rural status**

```r
table(lca_pcp_2009_12423$urban_rural)
```

```

Large Rural Small Rural       Urban 
       1202         497        7000 
```

**Female**

```r
table(lca_pcp_2009_12423$female)
```

```

   0    1 
5266 3433 
```

**Racial/ethnic identity**

Race/ethnicity does appear to be already classified as mutually exclusive

```r
table(lca_pcp_2009_12423$white,lca_pcp_2009_12423$hispanicrace)
```

```
   
       0    1
  0 2139  304
  1 6256    0
```

Checking with other race/ethnicity groups.

```r
table(lca_pcp_2009_12423$white,lca_pcp_2009_12423$black)
```

```
       0    1
  0 1518  925
  1 6256    0
```

```r
table(lca_pcp_2009_12423$race_ethnicity)
```

```
American Indian/Alaskan Native         Asian/Pacific Islander 
                            60                            859 
                      Hispanic             Non-Hispanic Black 
                           304                            925 
            Non-Hispanic White                          Other 
                          6256                            257 
                       Unknown 
                            38 
```

### Generational Cohort

```r
table(lca_pcp_2009_12423$generational)
```

```
 Baby Boomer, Leading Edge (1946-1954) Baby Boomer, Trailing Edge (1955-1964) 
                                  1700                                   2461 
                    Gen X (1965-1980)                      Gen Y (1981-2000) 
                                  3867                                      9 
                     WWII (1928-1945) 
                                   662 
```

### Average years of practice

There does appear to be 18 individuals missing practice years

```r
sum(is.na(lca_pcp_2009_12423$practice_yrs))
```

```
[1] 18
```

Now looking at average years of practice among those with non-missing values

```r
mean(lca_pcp_2009_12423$practice_yrs, na.rm = TRUE)
```

```
[1] 19.95358
```

**Average age**

No individuals missing values for age

```r
sum(is.na(lca_pcp_2009_12423$age))
```

```
[1] 0
```

Now looking at average years of practice among those with non-missing values

```r
mean(lca_pcp_2009_12423$age, na.rm = TRUE)
```

```
[1] 46.50213
```

**Medical school**

Seem we consider IMG medical school vs. other

```r
table(lca_pcp_2009_12423$school_img)
```

```

   0    1 
7196 1455 
```

### Looking at the 2019 dataset

Note that the variables in the 2009 dataset are the exact same variables in the 2019 dataset.

There are a total of 7,926 individuals in the 2019 dataset.

```r
lca_pcp2019_12423   <- read.table("Data/lca_pcp2019_1.24.23.dat",
                                  header = TRUE, sep = ",", skip = 2)
#Changing all values of -9999 to missing
lca_pcp2019_12423[lca_pcp2019_12423==-9999] <- NA

names(lca_pcp2019_12423) <- vars_2009
```

Now we need to format the variables to be categorical variables

```
lca_pcp2019_12423 <- lca_pcp2019_12423 %>%
  mutate(specialty = case_when(
    group_fam   == 1 ~ "Family Medicine",
    group_int   == 1 ~ "General Internal Medicine",
    group_obgyn == 1 ~ "General Obstetrics and Gynecology",
    group_ped   == 1 ~ "General Pediatrics",
    group_ger   == 1 ~ "Geriatrics",
    TRUE             ~ "Missing"
  ),
  employment_setting = case_when(
    setting_amb   == 1 ~ "Ambulatory Care",
    setting_hosp  == 1 ~ "Hospital",
    setting_medsch == 1 ~ "Medical School/Parent University",
    setting_nurs  == 1 ~ "Nursing Home/Extended Care",
    setting_other == 1 ~ "Other",
    setting_solo == 1 ~ "Solo practioner",
    setting_missing == 1 ~ "Missing",
    TRUE             ~ "Missing Value"
  ),
  urban_rural = case_when(
    ruca_urban  == 1 ~ "Urban",
    ruca_lrural == 1 ~ "Large Rural",
    ruca_srural == 1 ~ "Small Rural",
    TRUE             ~ "Missing"
  ),
  race_ethnicity = case_when(
    white == 1           ~ "Non-Hispanic White",
    black == 1           ~ "Non-Hispanic Black",
    american_native == 1 ~ "American Indian/Alaskan Native",
    asian_or_pi == 1     ~ "Asian/Pacific Islander",
    hispanicrace == 1    ~ "Hispanic",
    other_race == 1      ~ "Other",
    race_unknown == 1    ~ "Unknown",
    TRUE                 ~ "Missing"
  ),
  generational = case_when(
    gen_x          == 1 ~ "Gen X (1965-1980)",
    gen_y          == 1 ~ "Gen Y (1981-2000)",
    gen_boom_lead  == 1 ~ "Baby Boomer, Leading Edge (1946-1954)",
    gen_boom_trail == 1 ~ "Baby Boomer, Trailing Edge (1955-1964)",
    gen_ww2        == 1 ~ "WWII (1928-1945)",
```

```r
    TRUE                  ~ "Missing"
  ))
```

### Specialty

We do have no individuals missing specialty in 2019, which aligns with descriptive table.

```r
table(lca_pcp2019_12423$specialty)
```

```
                 Family Medicine         General Internal Medicine 
                            2825                              2062 
General Obstetrics and Gynecology                General Pediatrics 
                            1160                              1629 
                      Geriatrics 
                             250 
```

### Employment Setting

We do have 572 individuals missing employment setting which aligns with Table 1, where we see some degree of missingness for employment setting

```r
table(lca_pcp2019_12423$employment_setting)
```

```
                 Ambulatory Care                          Hospital 
                            4305                              1443 
Medical School/Parent University                           Missing 
                             240                               572 
      Nursing Home/Extended Care                             Other 
                             123                               678 
                 Solo practioner 
                             565 
```

How would we want to handle missing categories? Drop them from the analysis?

### Practice Characteristics

While other variables are mutually exclusive providing routine obstetric care and providing routine prenatal care appear to be not mutually exclusive.

```r
table(lca_pcp2019_12423$prenatal,lca_pcp2019_12423$obstetrics)
```

```
       0    1
  0 6430   35
  1  251  916
```

### Average patient-care hours per week

Check on number of missing values for average patient-care hours per week - we have a total of 698 individuals missing.

```r
sum(is.na(lca_pcp2019_12423$patient_care_hours))
```

```
[1] 698
```

Looking at descriptive statistics by patient care

```r
#Mean for patient care hours
mean(lca_pcp2019_12423$patient_care_hours, na.rm=TRUE)
```

```
[1] 38.38046
```

```r
#Mean for Standardized values of Y2009_patient_care_hours
mean(lca_pcp2019_12423$zhours, na.rm=TRUE)
```

```
[1] -0.0002234069
```

### Urban/rural status

We do have 11 individuals missing Urban/Rural in 2019

```r
table(lca_pcp2019_12423$urban_rural)
```

```
Large Rural     Missing Small Rural       Urban
        944          11         367        6604
```

### Female

```r
table(lca_pcp2019_12423$female)
```

```
   0    1 
3957 3969 
```

**Racial/ethnic identity**

Can't seem to identify the 3 individuals missing - unless they are captured within the 267 unknown?

```r
table(lca_pcp2019_12423$race_ethnicity)
```

```
American Indian/Alaskan Native         Asian/Pacific Islander 
                            48                            835 
                      Hispanic             Non-Hispanic Black 
                           321                            899 
            Non-Hispanic White                          Other 
                          5349                            207 
                       Unknown 
                           267 
```

**Generational Cohort**

```r
table(lca_pcp2019_12423$generational)
```

```
 Baby Boomer, Leading Edge (1946-1954) Baby Boomer, Trailing Edge (1955-1964) 
                                   911                                   1790 
                     Gen X (1965-1980)                      Gen Y (1981-2000) 
                                  3340                                   1686 
                      WWII (1928-1945) 
                                   199 
```

**Average years of practice**

There does appear to be 4 individuals missing practice years

```r
sum(is.na(lca_pcp2019_12423$practice_yrs))
```

```
[1] 4
```

Now looking at average years of practice among those with non-missing values

```r
mean(lca_pcp2019_12423$practice_yrs, na.rm = TRUE)
```

```
[1] 22.46541
```

#### Average age

No individuals missing values for age

```r
sum(is.na(lca_pcp2019_12423$age))
```

```
[1] 0
```

Now looking at average years of practice among those with non-missing values

```r
mean(lca_pcp2019_12423$age, na.rm = TRUE)
```

```
[1] 49.22117
```

#### Medical school

Seem we consider IMG medical school vs. other

```r
table(lca_pcp2019_12423$school_img)
```

```

   0    1 
6709 1214 
```

### Creating combined datasets

Changing variable names to _2009 or _2019 depending on what dataset they are from

Now merging the dataset keeping only individuals in both datasets

```r
merged_lta <- merge(lca_pcp_2009_12423, lca_pcp2019_12423, by = "indexno")
```

This results in a total of 4,614 individuals. Could we look at descriptive statistics for combinations of variables across time

In particular we are not sure about specialty, given questions have changed over time so this shifted broad patterns of what specialties show up. Not necessarily a change of practice, but a change in how they reported their practice.

### Descriptive stats for people in 2009 dataset only

### Descriptive stats for people in 2019 dataset only

Getting from the 2009

### Now looking at descriptive stats for speciality across years

```r
table(merged_lta$specialty_2009,merged_lta$specialty_2019)
```

```
                                 Family Medicine General Internal Medicine
  Family Medicine                           1619                         3
  General Internal Medicine                   22                      1156
  General Obstetrics and Gynecology            8                         1
  General Pediatrics                           6                         6
  Geriatrics                                   1                         6
  Missing                                      0                         1

                                 General Obstetrics and Gynecology
  Family Medicine                                                4
  General Internal Medicine                                      0
  General Obstetrics and Gynecology                            660
  General Pediatrics                                             0
  Geriatrics                                                     0
  Missing                                                        0

                                 General Pediatrics Geriatrics
  Family Medicine                                 9         77
  General Internal Medicine                      40         50
```

```
General Obstetrics and Gynecology                      0          0
General Pediatrics                                   919          1
Geriatrics                                             0         23
Missing                                                1          1
```

```r
table(merged_lta$employment_setting_2009,merged_lta$employment_setting_2019)
```

```

                                Ambulatory Care Hospital
Ambulatory Care                            2320      208
Hospital                                    135      204
Medical School/Parent University             26       31
Missing                                     132       30
Nursing Home/Extended Care                    2        0
Other                                        14        5
Solo practioner                             214       24

                                Medical School/Parent University Missing
Ambulatory Care                                               15     124
Hospital                                                      10      25
Medical School/Parent University                             102       3
Missing                                                        9      15
Nursing Home/Extended Care                                     0       1
Other                                                          0       2
Solo practioner                                                2      19

                                Nursing Home/Extended Care Other
Ambulatory Care                                         29   207
Hospital                                                 8    86
Medical School/Parent University                         1    10
Missing                                                  3    25
Nursing Home/Extended Care                              32     7
Other                                                    0    10
Solo practioner                                          4    37

                                Solo practioner
Ambulatory Care                             133
Hospital                                     21
Medical School/Parent University              1
Missing                                      23
Nursing Home/Extended Care                    0
Other                                         2
```

```
  Solo practioner                                   303
```

```r
table(merged_lta$urban_rural_2009,merged_lta$urban_rural_2019)
```

```
               Large Rural Missing Small Rural Urban
  Large Rural          512       0          21   108
  Small Rural           22       0         185    46
  Urban                 62       1          31  3626
```